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AI Lab

Week - 1

1. Particle Swarm Optimization:

Objective: Minimize Multimodal Rastrigin Function

```
import random
import math

def rastrigin(x, y):
    return 20 + (x**2) + (y**2) - 10 * math.cos(2 * math.pi * x) - 10 * math.cos(2 * math.pi * y)

def execute_pso(w, c1, c2, num_particles=30, max_iterations=100, bound=5.12):
    particles = []

    for _ in range(num_particles):
        x = random.uniform(-bound, bound)
        y = random.uniform(-bound, bound)
        vx = random.uniform(-1, 1)
        vy = random.uniform(-1, 1)
        fitness = rastrigin(x, y)

        particles.append([(x, y), (vx, vy), (x, y), fitness])

    global_best_pos = None
    global_best_fitness = float('inf')

    for p in particles:
        if p[3] < global_best_fitness:
            global_best_fitness = p[3]
            global_best_pos = p[2]

    for _ in range(max_iterations):

        for p in particles:
            x, y = p[0]
            vx, vy = p[1]
            px, py = p[2]

            r1x, r1y = random.random(), random.random()
            r2x, r2y = random.random(), random.random()

            new_vx = (
                w * vx
                + c1 * r1x * (px - x)
                + c2 * r2x * (global_best_pos[0] - x)
            )

            new_vy = (
                w * vy
                + c1 * r1y * (py - y)
                + c2 * r2y * (global_best_pos[1] - y)
```

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)

new_x = x + new_vx
new_y = y + new_vy

new_x = max(-bound, min(bound, new_x))
new_y = max(-bound, min(bound, new_y))

new_fitness = rastrigin(new_x, new_y)

if new_fitness < p[3]:
    p[2] = (new_x, new_y)
    p[3] = new_fitness

p[0] = (new_x, new_y)
p[1] = (new_vx, new_vy)

for p in particles:
    if p[3] < global_best_fitness:
        global_best_fitness = p[3]
        global_best_pos = p[2]

return global_best_fitness

def get_mean_fitness(w, c1, c2, runs=5):
    total_fitness = sum(
        execute_pso(w, c1, c2)
        for _ in range(runs)
    )
    return total_fitness / runs

print("Objective: Minimize Multimodal Rastrigin Function\n")

print("--- Table 1: Effect of inertia weight w ---")
print(f"{'Case':<15} | {'w':<4} | {'c1':<4} | {'c2':<4} | {'Mean Best Fitness'}")
print("-" * 55)

for case, w in [
    ("Low inertia", 0.4),
    ("Medium inertia", 0.7),
    ("High inertia", 0.9)
]:
    mean_fit = get_mean_fitness(w, 2.0, 2.0)

print(
    f"{'case':<15} | {'w':<4} | {'2.0':<4} | {'2.0':<4} | {'mean_fit:.6f}'"

```

```

)

print("\n--- Table 2: Effect of cognitive coefficient c1 ---")
print(f"{'Case':<15} | {'w':<4} | {'c1':<4} | {'c2':<4} | {'Mean Best Fitness'}")
print("-" * 55)

for case, c1 in [
    ("Low c1", 0.5),
    ("Medium c1", 1.5),
    ("High c1", 2.5)
]:
    mean_fit = get_mean_fitness(0.7, c1, 2.0)

    print(
        f"{'case':<15} | {'0.7':<4} | {'c1':<4} | {'2.0':<4} | {'mean_fit:.6f}'"
    )

print("\n--- Table 3: Effect of social coefficient c2 ---")
print(f"{'Case':<15} | {'w':<4} | {'c1':<4} | {'c2':<4} | {'Mean Best Fitness'}")
print("-" * 55)

for case, c2 in [
    ("Low c2", 0.5),
    ("Medium c2", 1.5),
    ("High c2", 2.5)
]:
    mean_fit = get_mean_fitness(0.7, 2.0, c2)

    print(
        f"{'case':<15} | {'0.7':<4} | {'2.0':<4} | {'c2':<4} | {'mean_fit:.6f}'"
    )

print("\n--- Table 4: Overall Comparison Table ---")
print(f"{'Case':<15} | {'w':<4} | {'c1':<4} | {'c2':<4} | {'Mean Best Fitness'}")
print("-" * 55)

experiments = [
    ("Baseline", 0.7, 2.0, 2.0),
    ("Low w", 0.4, 2.0, 2.0),
    ("High w", 0.9, 2.0, 2.0),
    ("Low c1", 0.7, 0.5, 2.0),
    ("High c1", 0.7, 2.5, 2.0),
    ("Low c2", 0.7, 2.0, 0.5),
    ("High c2", 0.7, 2.0, 2.5)
]

```

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]

for case, w, c1, c2 in experiments:
    mean_fit = get_mean_fitness(w, c1, c2)

    print(
        f"{case:<15} | {w:<4} | {c1:<4} | {c2:<4} | {mean_fit:.6f}"
    )

    print(
        "\nObservation: Good performance requires balancing exploration "
        "(higher w) and exploitation (lower w). The results may vary "
        "slightly per execution because PSO uses random numbers."
    )
```

I/O:

Output – 1:

Objective: Minimize Multimodal Rastrigin Function

--- Table 1: Effect of inertia weight w ---

Case	w	c1	c2	Mean Best Fitness
Low inertia	0.4	2.0	2.0	0.000000
Medium inertia	0.7	2.0	2.0	0.000006
High inertia	0.9	2.0	2.0	0.576904

--- Table 2: Effect of cognitive coefficient c1 ---

Case	w	c1	c2	Mean Best Fitness
Low c1	0.7	0.5	2.0	0.198992
Medium c1	0.7	1.5	2.0	0.000001
High c1	0.7	2.5	2.0	0.001807

--- Table 3: Effect of social coefficient c2 ---

Case	w	c1	c2	Mean Best Fitness
Low c2	0.7	2.0	0.5	0.000000
Medium c2	0.7	2.0	1.5	0.000000
High c2	0.7	2.0	2.5	0.203526

--- Table 4: Overall Comparison Table ---

Case	w	c1	c2	Mean Best Fitness
Baseline	0.7	2.0	2.0	0.000521
Low w	0.4	2.0	2.0	0.000000
High w	0.9	2.0	2.0	0.311580
Low c1	0.7	0.5	2.0	0.198992
High c1	0.7	2.5	2.0	0.045740
Low c2	0.7	2.0	0.5	0.000000
High c2	0.7	2.0	2.5	0.033613

Observation: Good performance requires balancing exploration (higher w) and exploitation (lower w). The results may vary slightly per execution because PSO uses random numbers.

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